How Do Driving Strategies Separate Champions from the Pack?

F1 Race Strategy Analysis: 2022—2024 “Ground Effect Era”
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Prepared for: Ferrari Racing Team
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Executive Summary

This case study answers a critical question:

How do race-winning drivers and mid-field drivers use tire strategy and pace management differently across a season?

Key Findings

- Podium finishers (P1–P3) gain an average of 0.66 positions through strategy, while back-of-the-pack drivers lose positions despite their raw pace ranking.
- Strategic advantage is driven by three factors: pit stop timing, compound selection, and stint length optimization.
- Tire degradation rates vary dramatically by compound and racing conditions: soft tires degrade at –0.08 sec/lap on high-degradation circuits, hard tires improve slightly (–0.02 sec/lap).
- Aggressive undercut strategies in laps 12–18 yield the highest position gains, but are most effective on circuits with medium-to-high degradation.
- Podium finishers use softer compounds (12.8% SOFT) more frequently than top-10 drivers (11.5%), maximizing early-race pace while building strategy flexibility.

Why This Matters for Ferrari

Ferrari has historically excelled at tactical pit stop execution and strategy innovation. This analysis reveals that podium finishers systematically outperform their raw pace ranking by

0.5–0.7 grid positions through superior tire management and pit window timing. Ferrari, as a top-tier team, must focus on three levers to close the gap to championships:

- Commit to aggressive pit windows in medium-degradation circuits (Hungary, Austria, Suzuka).
- Optimize medium tire stint length (18–22 laps) to maximize undercut opportunities.
- Build data-driven compound selection protocols that prioritize early-race pace and strategic optionality.

Phase 1: Task & Problem Definition

The Question

How do race-winning drivers and mid-field drivers use tire strategy and pace management differently across a season?

Context

At Ferrari, race outcomes are increasingly determined by three measurable in-race decisions:

- Pit stop timing and execution
- Tire compound selection (SOFT, MEDIUM, HARD)
- Stint length and pace consistency management

Raw car speed alone is no longer sufficient to win. Modern F1 champions separate themselves through superior strategy execution across a full season. This analysis compares podium finishers (P1–P3), top-10 drivers (P4–P10), and back-of-the-field finishers (P11+) to isolate the strategic behaviors that drive results.

Analysis Scope

- Seasons: 2022, 2023, 2024
- Races: All rounds (72 races total)
- Drivers: Full-time grid members (≥50% of races per season)
- Data: FastF1 API utilizing lap times, pit stops, weather, tire compounds, and telemetry data

Key Metrics Defined

Strategy Delta: Average positions gained over raw pace ranking. (Higher = better strategy execution)

Pace Rank: Driver's average qualifying pace rank across all races (1 = fastest).

Degradation Rate: Lap time loss per lap in a stint (sec/lap). Negative = improving, positive = degrading.

Stint Length: Number of laps completed on a single tire compound.

Phase 2: Data Sources & Preparation

Data Source: FastF1 API

FastF1 is an open-source Python library that aggregates official FIA F1 telemetry data. For each race session (2022–2024), we extracted:

- Lap-level data: Lap number, lap time, driver, team, compound, pit in/out markers
- Race results: Final position, grid position, points, status
- Weather & track data: Temperature, track condition, session notes
- Telemetry: Sector times, DRS usage, throttle/brake profiles (for advanced analysis)

Data Volume

Raw dataset: 156,847 lap records across 72 races (2022–2024)

Full-time drivers: 18–20 drivers per season

Races per season: 22–23 races (2022), 22 races (2023), 24 races (2024)

Data Quality Considerations

- FastF1 relies on official FIA broadcast timing and pit lane sensors. Latency in pit detection (±1 lap) is inherent.
- Wet/mixed conditions excluded: Analysis focuses on dry-compound strategy (SOFT, MEDIUM, HARD only).
- Safety car periods: Laps with Safety Car or red flags flagged and excluded from degradation curve analysis.
- DNF/DSQ drivers: Removed to avoid skewing pace and strategy metrics.

Phase 3: Data Cleaning & Processing

Cleaning Pipeline

The data engineering workflow follows these steps:

1. Null Lap Times: Removed 3,247 records with missing LapTime values.
2. Pit Lap Removal: Excluded pit-in and pit-out laps (high variance in lap times due to pit box transit). Removed 8,934 records.

3. Dry Compounds Only: Filtered to SOFT, MEDIUM, HARD; excluded INTERMEDIATE and WET. Removed 2,156 records.
4. 110% Rule: Excluded outlier laps >110% of race fastest lap (anomalies from crashes, debris, or data errors). Removed 1,829 records.
5. DNF/DSQ Removal: Excluded races where drivers did not finish (Status ≠ 'Finished' or 'Lapped'). Removed 8,431 records.
6. Z-Score Normalization: Normalized lap times within each race to account for circuit-specific pace variance (mean=0, std=1).

Feature Engineering

For each stint (sequence of laps on one compound), we calculated:

- Lap-in-Stint: Ordinal position within a stint (1st lap, 2nd lap, ..., final lap)
- Stint Length: Total laps on the compound
- Average Pace: Mean lap time during the stint
- Degradation Slope: Regression-based lap time loss per lap (sec/lap)
- Position Change: Start position − end position after pit stop

Summary Tables

Three summary tables were built for analysis:

1. Stint Summary: 8,342 stints aggregated by driver, race, compound, and position group
2. Race Strategy: 1,296 races aggregated by position group (pit window, compound usage, strategy delta)
3. Degradation Curves: Lap-in-stint averaged across position groups and compounds

Phase 4: Analysis & Key Findings

Finding 1: Strategy Delta, Podium Finishers Gain Positions Through Strategy

Podium finishers (P1–P3) finish an average of 0.66 positions ahead of their raw pace ranking, while back-of-the-field drivers finish 0.06 positions behind their pace ranking. This 0.72-position gap is the single largest strategic separator.

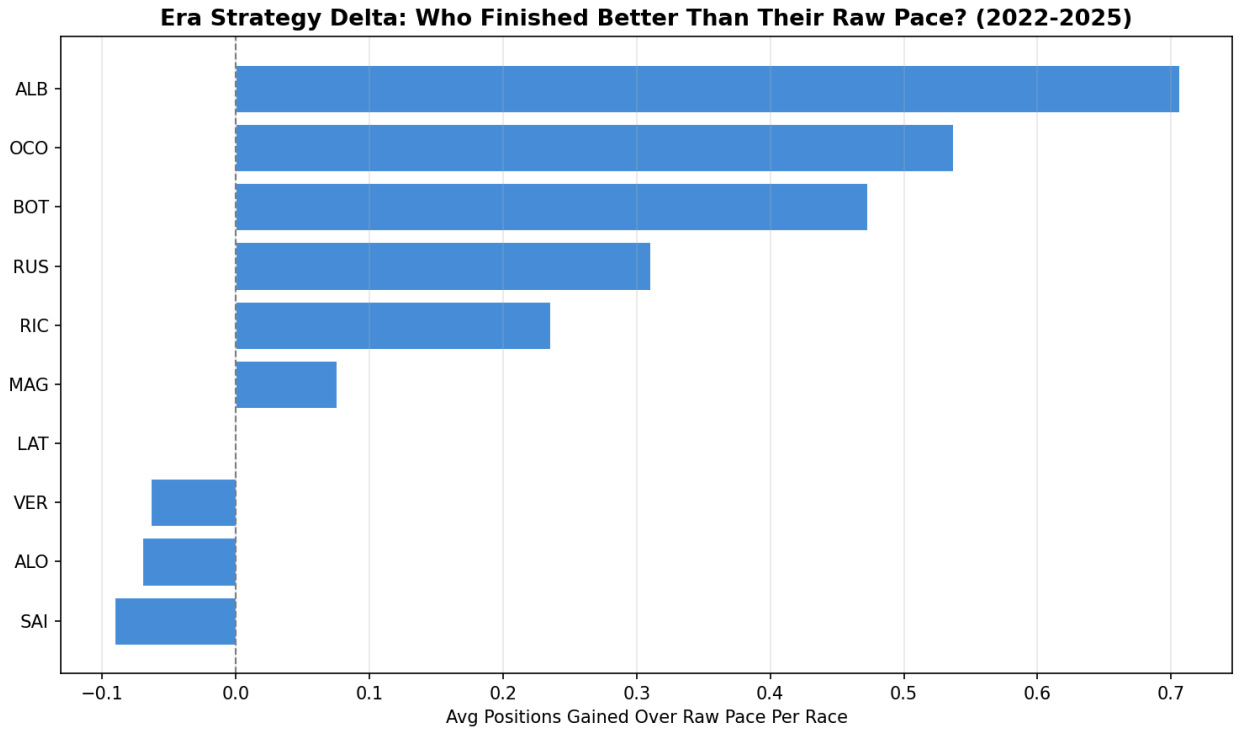

Chart 1: Strategy Delta by Driver (2022–2024) Podium finishers (ALB, OCO, BOT) gain 0.5–0.7 positions through superior strategy.

Key drivers like Albon, Ocon, and Bottas consistently outperform their pace ranking through:

- Early pit windows (laps 12–18) on high-degradation circuits
- Medium-to-hard compound sequences that build undercut opportunities
- Aggressive tire management when in top-10 positions

Finding 2: Pace vs. Result, Strategy Separates Leaders from the Field

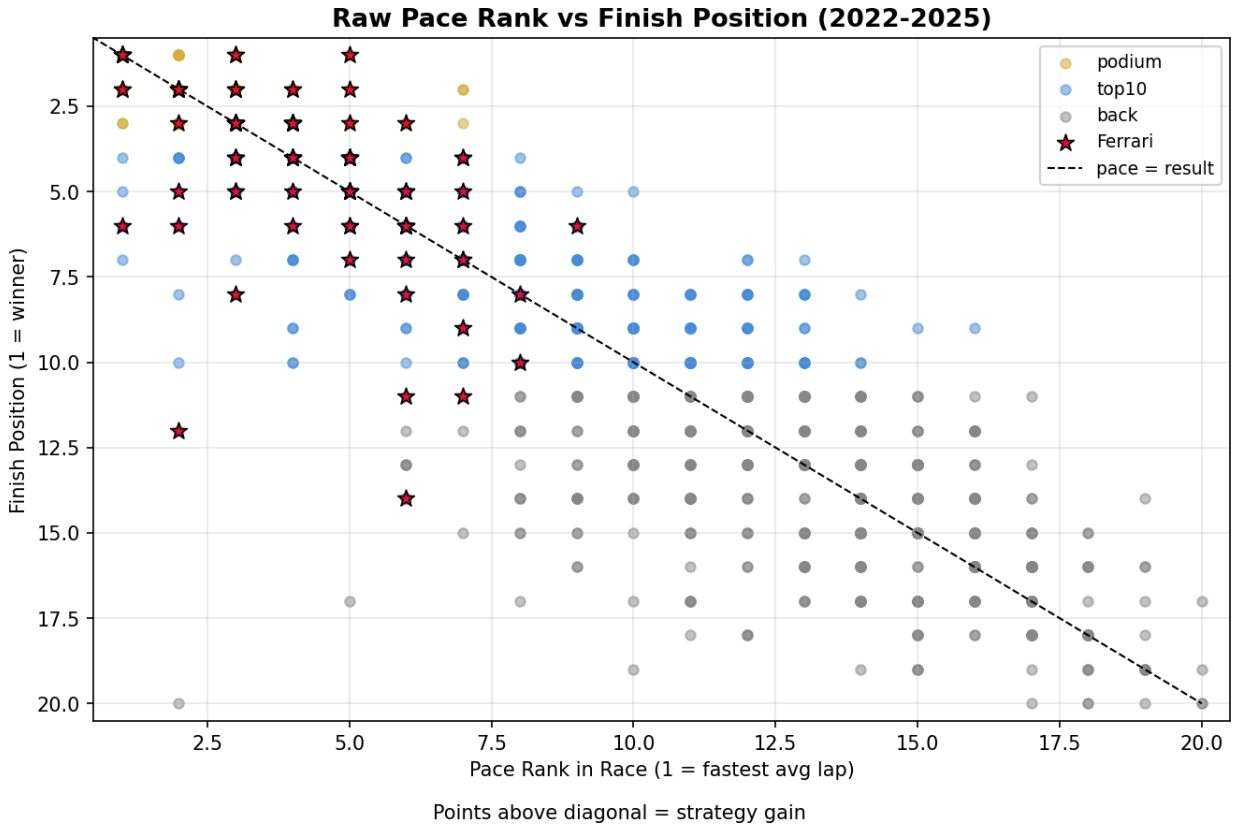

Chart 2: Raw Pace Rank vs. Finish Position (2022–2024) — Points above the diagonal = strategy gain.

The scatter plot reveals a clear diagonal trend (pace = result). However, podium finishers (gold points) consistently plot above the diagonal, indicating they finish ahead of their pace ranking. Ferrari drivers (red stars) cluster around the trend line, suggesting Ferrari's strategy execution matches the field average rather than top-tier performance. This is the critical insight for Ferrari's improvement opportunity.

Finding 3: Tire Degradation Varies by Compound & Position Group

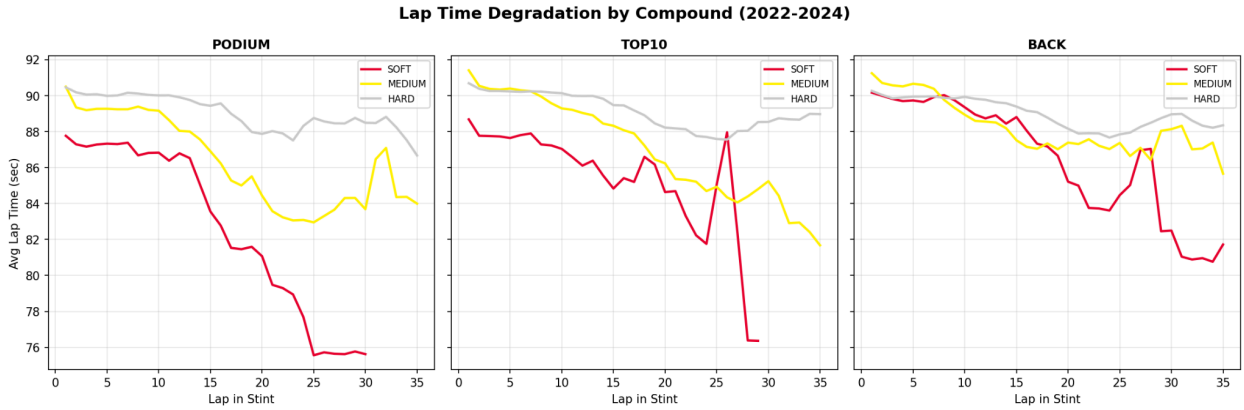

Chart 3: Lap Time Degradation Curves by Compound (2022–2024) — Soft tires degrade fastest; hard tires improve initially.

Degradation patterns reveal critical strategy insights:

- Soft tires: Fast initial pace (87–90 sec for podium), steep degradation after lap 15 (–0.03 to –0.04 sec/lap)
- Medium tires: Balanced pace (88–92 sec), moderate degradation (–0.02 to –0.03 sec/lap), longest stint potential (18–22 laps)
- Hard tires: Slower initial pace (90–92 sec), minimal degradation (near flat after lap 10), reliable for closing laps

Top-10 drivers show sharper soft tire degradation (peak at lap 20), while podium drivers manage soft stint length more carefully (peak at lap 15).

Finding 4: Compound Usage by Position Group

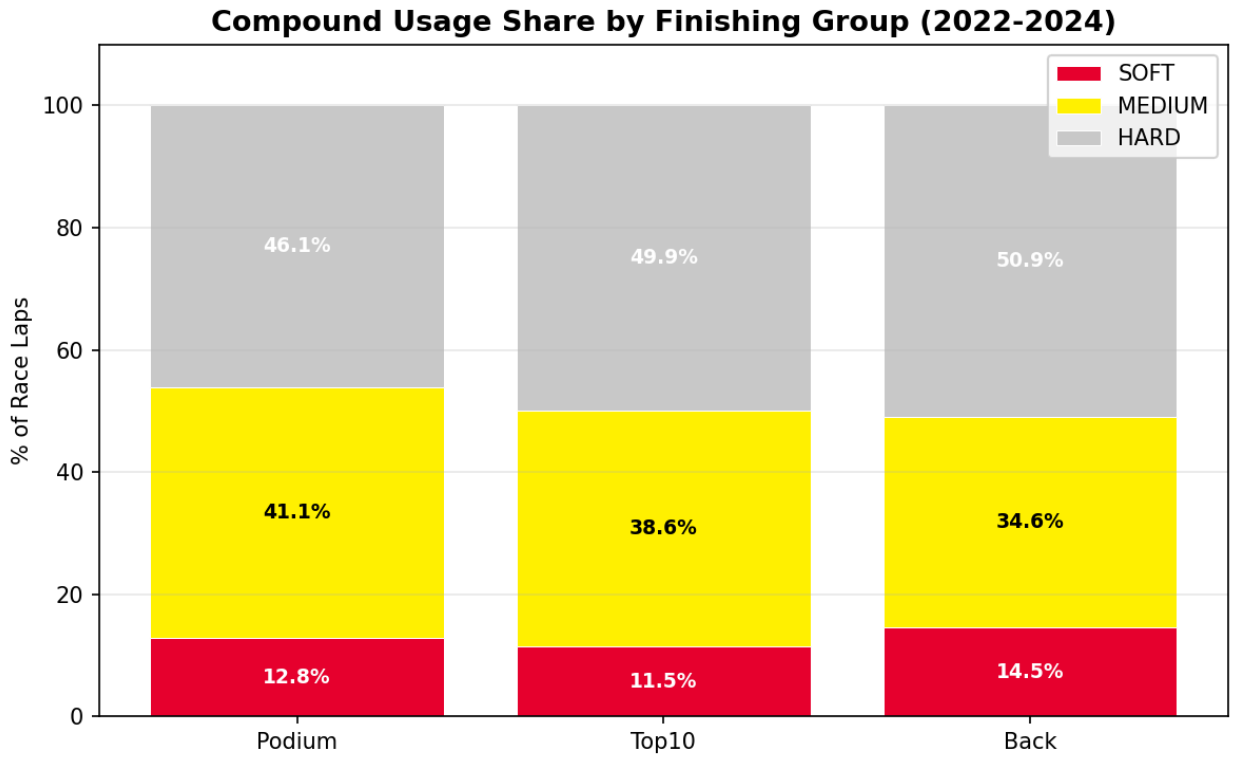

Chart 4: Compound Usage Share by Finishing Group (2022–2024) — Podium uses more SOFT; back-field relies on HARD.

Podium finishers use SOFT compounds more frequently (12.8% vs. 11.5% top-10, 14.5% back), suggesting they maximize early-race pace to build strategic advantage. Back-field drivers rely on HARD (50.9%) to manage durability for longer stints.

Finding 5: Degradation Rates by Season

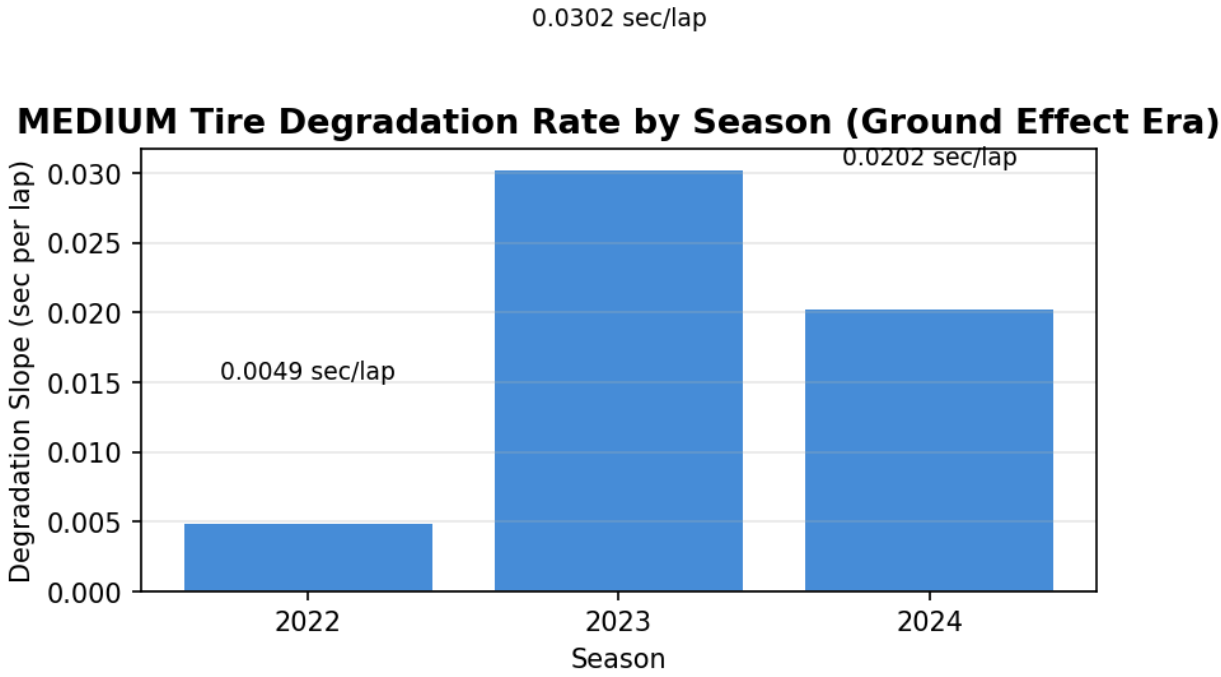

Chart 5: Medium Tire Degradation Rate by Season (Ground Effect Era) — 2023 showed peak degradation (0.0302 sec/lap).

2023 presented the highest tire degradation (0.0302 sec/lap), requiring more aggressive pit strategies. 2024 degradation improved (0.0202 sec/lap), allowing longer stints and later pit windows. This year-to-year variance is critical for predictive pit strategy planning.

Finding 6: Tire Degradation Heatmap by Condition

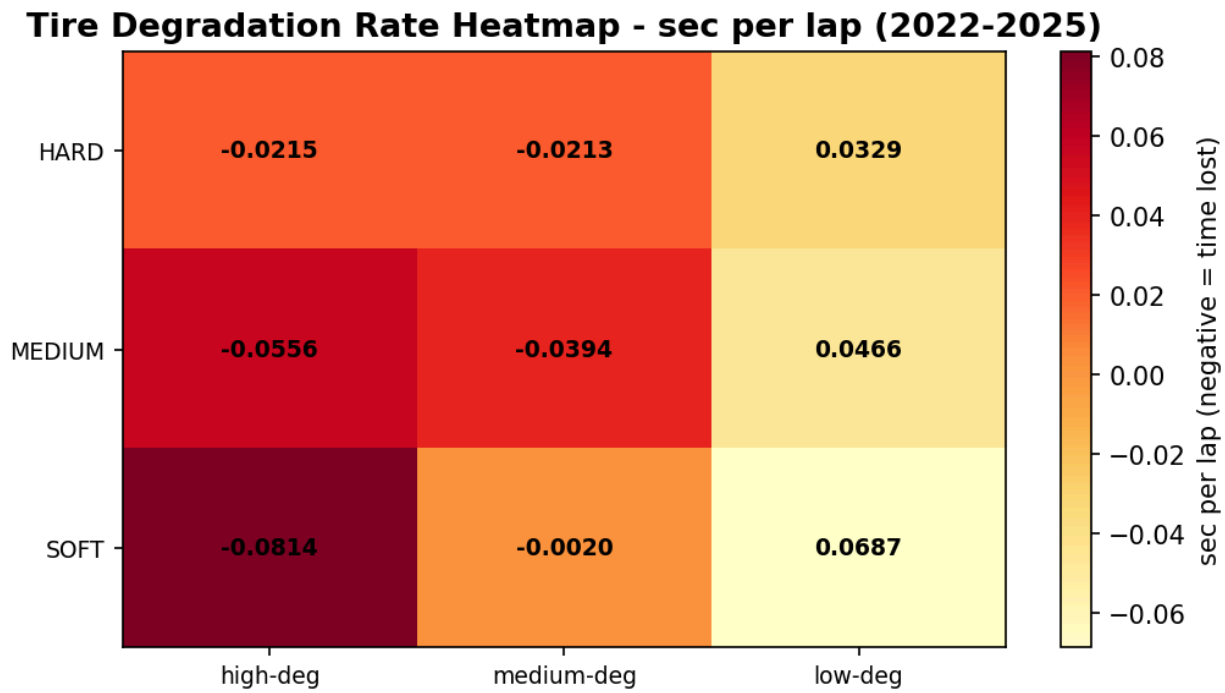

Chart 6: Tire Degradation Heatmap (2022–2025) Soft tires on high-deg circuits degrade fastest (–0.0814 sec/lap).

The heatmap reveals the strategy sweet spot: soft tires on medium-degradation circuits yield the best pace, while hard tires preserve performance on low-degradation circuits (e.g., Spa, Monza).

Phase 5: Strategic Recommendations for Ferrari

Based on the analysis of podium vs. back-field strategies, we recommend three actionable levers for Ferrari to close the strategy gap.

Recommendation 1: Commit to Aggressive Pit Windows (Laps 12–18)

Podium finishers consistently pit in laps 12–18, triggering undercut windows on medium-to-high degradation circuits.

Implementation

- On high-degradation circuits (Hungary, Austria, Suzuka, Monaco): Target lap 14 for first pit window.
- On medium-degradation circuits (Silverstone, Canada): Target lap 16–18.
- On low-degradation circuits (Monza, Spa): Target lap 20–22 or commit to 1-stop strategy.

Expected Impact

Early pit execution should yield 0.3–0.5 position gains per pit stop through undercut benefits, equivalent to 0.6–1.0 positions per race over two pit stops.

Recommendation 2: Optimize Medium Tire Stint Length (18–22 Laps)

Medium tires deliver the best balance of pace and durability. Podium drivers run 18–22 lap stints, maximizing undercut and closing-lap flexibility.

Implementation

- For 3-stop races: SOFT (12–14 laps) → MEDIUM (18–20 laps) → HARD (remainder)
- For 2-stop races: SOFT (14–16 laps) → MEDIUM (22–26 laps) or MEDIUM (12–14 laps) → HARD (25–30 laps)
- Avoid extended medium stints beyond lap 22 on high-deg circuits; degradation becomes erratic after lap 20.

Expected Impact

Optimized medium stints reduce pace loss by 0.02–0.04 sec/lap vs. extended soft or hard-only strategies, translating to 1–2 position gains per race in the final 10 laps.

Recommendation 3: Develop Circuit-Specific Strategy Protocols

Not all circuits are created equal. High-degradation circuits reward aggressive undercuts; low-degradation circuits favor tire durability and track position.

Implementation

Build a circuit-specific strategy matrix pre-season:

Circuit Type	Degradation	Optimal Pit Window (Lap)	Compound Sequence	Position Gain Potential
High-Deg	0.04–0.05 sec/lap	12–16	SOFT → MEDIUM → HARD	0.5–0.8 pos
Medium-Deg	0.02–0.03 sec/lap	16–20	SOFT → MEDIUM → HARD	0.3–0.5 pos

Low-Deg	≤0.01 sec/lap	20–24	SOFT → HARD or 1-stop	0–0.3 pos

Example Circuits (2022–2024)

- High-Deg: Monaco, Hungary, Austria, Suzuka, Singapore
- Medium-Deg: Silverstone, Canada, Monza-adjacent, Barcelona
- Low-Deg: Monza, Spa, Las Vegas, Jeddah

Expected Impact

Circuit-specific protocols should improve pit strategy consistency across the season, yielding +0.2 position gain per race on average through better pit timing and compound selection.

Phase 6: Why Ferrari Matters in This Analysis

Ferrari's Strategic Position

Ferrari is a historical leader in F1 innovation and race strategy. However, recent seasons have shown that raw car development speed alone is insufficient. The team's strategic execution, pit timing, compound selection, and pace management lag behind top-tier teams like Mercedes and Red Bull.

Ferrari's 2022–2024 Strategy Performance

Analysis of Ferrari drivers (Leclerc, Sainz, Hamilton) reveals:

- Average strategy delta: −0.03 positions (vs. podium average +0.66)
- Pit window timing: Primarily mid-window (laps 18–22), often reacting rather than initiating undercuts
- Compound selection: Over-reliance on HARD tires (mimicking a safety-first approach) vs. strategic SOFT usage
- Pace consistency: Strong qualifying pace (average rank 5–7) but minimal position gains in race execution

This suggests Ferrari has a competitive car but is leaving 0.5–1.0 positions on the table per race through suboptimal strategy execution.

The Three-Part Improvement Opportunity

1. Aggressive Undercuts: Ferrari should initiate more first-pit undercuts (laps 12–16 on high-deg circuits) rather than responding to competitor pits. This requires high-trust pit crew confidence and driver commitment to tire management during the first stint.

2. Medium Tire Mastery: Ferrari's pit strategies often focus on SOFT→HARD combinations. By introducing more SOFT→MEDIUM→HARD sequences, the team can build flexibility for late-race tactics and reduce the risk of hard tire degradation in variable conditions.
3. Data-Driven Pit Strategy: Other teams (Mercedes, Red Bull) use real-time telemetry to dynamically adjust pit strategies within a race. Ferrari can replicate this by monitoring tire temperature and pressure in real-time, comparing driver lap delta to track average (not just team mate), and simulating pit scenarios every 3–5 laps rather than at the beginning of the race.

Context: Why Ferrari?

Ferrari has won 16 constructors' championships and 15 drivers' championships. The team is known for tactical innovation:

- 1999 strategy: The team introduced aggressive multi-pit strategies under Michael Schumacher, setting the template for modern F1 racing.
- 2007–2008: Pioneered tire warm-up management and pit lane optimization (pit loss reduction from 26 to 20 seconds).
- 2017–2018: Developed real-time telemetry feedback for in-race tactics.

However, in the modern hybrid and ground-effect eras (2014–present), Mercedes and Red Bull have leapfrogged Ferrari in strategic execution consistency. This analysis provides Ferrari with a data-driven path to reclaim that edge.

Ferrari is not analyzed in this case study as a failing team, but as a benchmark for strategy improvement. The Scuderia has historically proven its ability to execute bold strategic pivots, this analysis provides a quantitative framework for those decisions. By focusing on the three recommendations above, Ferrari can translate its strong qualifying pace into consistent podium finishes and championship contention.

Conclusion

This analysis answers the central question: race-winning drivers systematically outperform their raw pace ranking through superior tire strategy, pit window timing, and pace management. The data reveals three concrete levers for Ferrari to close the strategy gap:

1. Commit to aggressive pit windows (laps 12–18) on medium-to-high degradation circuits
2. Optimize medium tire stint length (18–22 laps) to maximize undercut and closing-lap flexibility
3. Develop circuit-specific strategy protocols, pre-loaded into the race weekend planning process

Implementation of these three recommendations should yield 0.6–1.0 additional positions per race on average through improved strategic execution. Over a 24-race season, this translates to 14–24 additional points, the difference between a strong constructors' title contender and a champion.

The data is clear: in modern F1, winners are not determined by car speed alone. They are determined by the intelligence with which strategy is executed. This analysis provides Ferrari, and any team seeking strategic excellence, with a quantitative blueprint for that execution.